# A Structural Reinterpretation of Dark Energy in Darke Theory

## Late-Time Acceleration as the Metric Appearance of Closure-Accessible Coherence Thinning

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**Abstract**

This paper develops a structural reinterpretation of dark energy within Darke Theory. The central claim is that late-time accelerated expansion does not, by itself, compel the ontology of a primitive dark-energy substance, vacuum-energy density, or independent expansive fluid. In Darke Theory, geometry is derivative rather than primitive. Effective cosmological equations therefore describe the readable large-scale projection of deeper structural evolution rather than the foundational ontology of the universe.

The paper proceeds in five stages. First, it places late-time homogeneous cosmology under the relativistic closure–curvature bridge, so that FLRW-type dynamics are treated as effective post-threshold projections rather than primitive laws. Second, it defines the relevant state variables as coarse-grained descendants of already established Darke-Theoretic quantities: closure density, accessible coherence, persistence burden, and future admissibility reserve. Third, it introduces a one-sided thinning operator for closure-accessible coherence, thereby ensuring that positive sourcing arises only from genuine thinning and not from unsigned change. Fourth, it formulates the effective cosmological sector in a two-level manner, separating the active acceleration source from the density sector entering the homogeneous constraint equation. This removes the continuity inconsistency present in weaker collapsed formulations. Fifth, it links the late-time acceleration source to the terminal-closure architecture of Darke Theory, in which admissible future reorganisation is progressively compressed under accumulated closure.

A provisional constitutive closure is then provided, not as a claim of final microscopic completion, but as a disciplined late-time bridge from structural variables to effective homogeneous stress-energy. The resulting framework shows how an observed acceleration history can be reproduced without requiring a primitive dark-energy ontology.

The strongest interpretation is therefore this: what is usually called dark energy may be the metric appearance of closure-accessible coherence thinning under increasing persistence burden and declining future admissibility reserve. Late-time acceleration is then not evidence for a hidden cosmic fluid, but a derivative sign that the universe is progressing toward terminal closure.

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# 1 Introduction

The standard dark-energy problem is usually posed within a pre-existing cosmological stage. One begins with a spacetime metric, adopts homogeneous cosmological equations, and then asks what additional component must be added to account for the observed late-time acceleration of the expansion history. The most common answer is that some dark sector exists whose effective pressure is sufficiently negative to drive acceleration.

Darke Theory begins earlier than this. It does not take geometry, spacetime, or fluid sectors as primitive. It begins with recursive differentiation and develops stable regimes only through stabilised refinement, closure, admissibility, and retained coherence. Geometry is a later readable encoding of sufficiently matured relation. A homogeneous cosmological metric is therefore already a derivative register, not a foundational ontology.

This changes the dark-energy question. If effective cosmological equations are derivative projections of deeper structure, then an acceleration term appearing in those equations need not originate in a primitive fluid or vacuum substance. It may instead be the readable large-scale signature of structural evolution within the underlying regime.

The purpose of this paper is to formulate that possibility precisely. The paper is not an anti-observational argument. The late-time acceleration data are taken seriously throughout. The issue is ontological underdetermination: the data may constrain an effective expansion history without uniquely fixing the underlying ontology.

The argument proceeds through five linked steps.

First, late-time cosmology is placed explicitly under the relativistic closure–curvature bridge. This prevents the silent reintroduction of homogeneous cosmology as a primitive axiom.

Second, the variables used in the late-time model are defined as coarse-grained descendants of already established structural objects in the Darke Theory corpus. They are not introduced as independent ad hoc cosmological substances.

Third, the source term is built from a one-sided thinning operator acting on closure-accessible coherence. This removes the sign ambiguity that would otherwise arise from using an unsigned magnitude of change.

Fourth, the effective cosmological dynamics are expressed in a two-level form. The active gravitational source for acceleration is distinguished from the effective density and pressure appearing in the homogeneous constraint equation. This separation is mathematically necessary.

Fifth, the model is placed inside the terminal-closure architecture. Late-time acceleration is interpreted as the derivative geometric appearance of narrowing futurehood under accumulated closure and rising persistence burden.

The resulting thesis is narrower than a full replacement cosmology but stronger than a mere interpretive analogy:

> the observational success of a dark-energy-like effective sector does not compel a primitive dark-energy ontology if a mathematically coherent structural lift exists.

## 2 Standard late-time cosmology and the usual ontological lift

In a homogeneous and isotropic late-time regime one writes

$$ds^2 = -dt^2 + a(t)^2 \left( \frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right), \tag{1}$$

with scale factor $a(t)$, spatial curvature $k$, and Hubble rate

$$H(t) := \frac{\dot{a}(t)}{a(t)}. \tag{2}$$

The effective homogeneous equations may be written as

$$H^2 + \frac{k}{a^2} = \frac{8\pi G}{3} \rho_{\text{eff}}, \tag{3}$$

and

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3} (\rho_{\text{eff}} + 3p_{\text{eff}}). \tag{4}$$

Acceleration requires

$$\rho_{\text{eff}} + 3p_{\text{eff}} < 0. \tag{5}$$

Since ordinary late-time matter and radiation do not provide this sign, one introduces a new effective component, commonly written with $w := p/\rho \approx -1$, and then lifts that effective sector to an ontology: cosmological constant, vacuum-energy density, dark fluid, quintessence, or related dark sector.

The critical point is that the data directly constrain only the effective homogeneous history. They do not, by themselves, uniquely determine the ontological lift from that history to an underlying primitive sector.

This paper asks whether an alternative lift exists within Darke Theory.

## 3 Darke-Theoretic foundation

Only the load-bearing pieces of the wider corpus are needed here.

### 3.1 Recursive differentiation, closure, and admissibility

Darke Theory begins from recursive differentiation rather than a completed inventory of objects. Stable persistence arises through refinement that does not dissolve itself. Closure marks the formation of self-bounding and self-maintaining structural organisation. Admissibility restricts the lawful continuation of structure: not every describable transition is structurally realisable.

This makes two points relevant for cosmology.

First, geometry is not primitive. It is a readable regime appearing only when structural maturity supports such a projection.

Second, the large-scale future of the universe is not fully described by metric kinematics alone. Closure history, coherence retention, burden, and admissible future reserve remain part of the load-bearing ontology.

### 3.2 Bridge principle

The relativistic closure–curvature bridge establishes that Lorentzian geometry becomes admissible only in a post-threshold regime and that homogeneous cosmology is therefore an effective projection of deeper structural dynamics. Homogeneous cosmological equations may be used in that regime, but they should not be treated as ontologically primitive.

This paper follows that principle strictly. Every FLRW-type equation below is interpreted as an effective late-time projection and not as the foundational description of reality.

**Remark 3.1.** The present paper does not derive the full universe from homogeneous cosmology. It derives an effective homogeneous late-time sector from a deeper structural ontology.

## 4 Inherited structural variables

The late-time model requires a small set of state variables. These are not introduced as new cosmological substances. They are coarse-grained cosmological descendants of structural quantities already established elsewhere in the theory.

**Definition 4.1** (Closure density)**.** The closure density

$$C(t) \tag{6}$$

is the coarse-grained large-scale measure of the extent to which existing structure constrains and organises its own admissible continuation.

**Definition 4.2** (Accessible coherence density)**.** The accessible coherence density

$$A(t) \tag{7}$$

is the coarse-grained density of coherence that remains structurally available for lawful future continuation, transport, and reorganisation.

**Definition 4.3** (Persistence burden)**.** The persistence burden

$$B(t) \tag{8}$$

is the effective carrying cost imposed by historically accumulated persistent structure.

**Definition 4.4** (Future admissibility reserve)**.** The future admissibility reserve

$$R(t) \tag{9}$$

is the coarse-grained measure of the remaining nontrivial lawful future reorganisation capacity available to the large-scale cosmic state.

### 4.1 Clarification of the closure variable $C(t)$

The role of $C(t)$ requires precision. Increased closure does not mean an increase in freedom. It means an increase in structurally integrated persistence. This has two aspects.

First, higher closure can indicate greater maturity and larger-scale integration of structure.

Second, that same increase can compress the admissible future by locking more of the system into already-formed organisation.

There is therefore no contradiction in allowing

$$\dot{C}(t) \geq 0 \tag{10}$$

while

$$\dot{R}(t) \leq 0. \tag{11}$$

A more closure-dense universe may be more internally integrated while also possessing less remaining lawful reorganisation capacity.

**Remark 4.5.** This distinction is important. Closure is an index of organised persistence, not a synonym for accessible novelty. The latter is encoded by $A$ and $R$, not by $C$ alone.

### 4.2 Closure-accessible coherence

The variable relevant to late-time acceleration is neither closure alone nor coherence alone. It is coherence as still available under the prevailing closure regime.

**Definition 4.6** (Closure-accessible coherence)**.** Define

$$\Xi(t) := C(t)A(t). \tag{12}$$

This quantity remains positive when coherent future structure is still available within a closure-bearing universe and declines when accessible future organisation thins.

### 4.3 Expected asymptotic tendencies

The late-time closure-dominant regime is naturally associated with the sign tendencies

$$\dot{C} \gtrsim 0, \qquad \dot{A} \lesssim 0, \qquad \dot{B} \gtrsim 0, \qquad \dot{R} \lesssim 0. \tag{13}$$

These are not universal monotonic axioms. They describe the expected asymptotic direction of a universe progressing toward terminal closure.

## 5 Dimensional discipline and regularity assumptions

The structural source must project into homogeneous cosmology with the correct dimensions. The effective acceleration source $D(t)$ must satisfy

$$[D] = T^{-2}. \tag{14}$$

A convenient choice is

$$[C] = [A] = [B] = [R] = 1, \tag{15}$$

with a coupling scale $\lambda$ carrying dimensions

$$[\lambda] = T^{-1}. \tag{16}$$

The remaining factor of inverse time arises from the thinning operator defined below.

The following regularity assumptions will be used throughout the late-time interval of interest.

**Assumption 5.1** (Regularity)**.** The functions $C, A, B, R$ are positive and piecewise $C^1$, and $R(t) + \varepsilon$ is bounded away from zero for each finite $t$, where $\varepsilon > 0$ is a regularising constant.

## 6 The one-sided thinning operator

An unsigned quantity such as

$$\left| \frac{\mathrm{d}}{\mathrm{d}t} \log \Xi \right|$$

is unsuitable for the present task because it assigns the same sign to thinning and growth. The structural interpretation pursued here requires a source only when closure-accessible coherence is genuinely thinning.

**Definition 6.1** (One-sided thinning operator)**.** For any positive piecewise $C^1$ scalar $Y(t)$, define

$$\Theta[Y](t) := \left[ -\frac{\mathrm{d}}{\mathrm{d}t} \log Y(t) \right]_+ = \max \left\{ 0,\ -\frac{\mathrm{d}}{\mathrm{d}t} \log Y(t) \right\}. \tag{17}$$

**Proposition 6.2** (Elementary properties)**.** *Under Assumption 5.1, the following hold for every positive piecewise $C^1$ scalar $Y$:*

*(i)* $\Theta[Y](t) \geq 0$*;*

*(ii) if* $\dot{Y}(t) \geq 0$*, then* $\Theta[Y](t) = 0$*;*

*(iii) if* $\dot{Y}(t) < 0$*, then* $\Theta[Y](t) = -\dot{Y}(t)/Y(t)$*.*

*Proof.* All three statements follow immediately from the definition. □

## 7 Darke Energy as a structural acceleration source

**Definition 7.1** (Darke Energy functional)**.** The effective structural acceleration source is

$$\boxed{D(t) = \lambda\, \Theta[\Xi](t)\, \frac{B(t)}{R(t) + \varepsilon}} \qquad \text{with} \qquad \Xi(t) = C(t)A(t), \tag{18}$$

where $\lambda > 0$ and $\varepsilon > 0$.

The interpretation is direct.

The factor $\Theta[\Xi]$ measures the rate at which closure-accessible coherence is being lost. The ratio $B/(R + \varepsilon)$ measures the structural pressure exerted by accumulated persistence burden relative

to the remaining reserve of admissible future reorganisation. The product therefore encodes a late-time source that is activated precisely when the universe is carrying increasing structural load while its available coherent future is thinning.

**Proposition 7.2** (Positivity and sign discipline)**.** *Under Assumption 5.1, $\lambda > 0$, and $\varepsilon > 0$,*

$$D(t) \geq 0. \tag{19}$$

*Moreover, $D(t) = 0$ whenever $\Xi(t)$ is nondecreasing.*

*Proof.* Immediate from (18) and the properties of $\Theta[\cdot]$. □

**Proposition 7.3** (Amplification under reserve compression)**.** *Holding $\Theta[\Xi]$ and $B$ fixed,*

$$\frac{\partial}{\partial R} D(t) < 0. \tag{20}$$

*Thus decreasing reserve amplifies the structural source.*

*Proof.* Differentiate the factor $(R + \varepsilon)^{-1}$. □

## 8 Admissibility of an effective homogeneous structural sector

The bridge principle permits homogeneous cosmology only as an effective post-threshold projection. Within that projection, the structural contribution must be represented by an effective homogeneous sector.

**Assumption 8.1** (Post-threshold homogeneous projection)**.** The late-time universe lies in a regime in which:

(i) Lorentzian geometry is an admissible projection;

(ii) a homogeneous coarse-grained description is valid at the level relevant for the late-time expansion history;

(iii) the structural variables $C, A, B, R$ are well-defined in that projection.

**Definition 8.2** (Effective structural sector)**.** Let

$$(\rho_{\mathrm{str}}(t), p_{\mathrm{str}}(t)) \tag{21}$$

denote the effective homogeneous density and pressure associated with the projected structural sector.

At this stage one must distinguish two different objects:

(a) the active gravitational source that appears in the acceleration equation;

(b) the density source that appears in the homogeneous constraint equation.

These are linked, but they are not identical by definition.

## 9 Effective homogeneous equations

Write ordinary late-time matter and radiation as

$$\rho_{\mathrm{mr}} := \rho_m + \rho_r, \qquad p_{\mathrm{mr}} := p_m + p_r. \tag{22}$$

The effective homogeneous equations are taken to be

$$\boxed{H^2 + \frac{k}{a^2} = \frac{8\pi G}{3}(\rho_{\mathrm{mr}} + \rho_{\mathrm{str}}),} \tag{23}$$

and

$$\boxed{\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho_{\mathrm{mr}} + 3p_{\mathrm{mr}}) + \frac{1}{3}D(t),} \tag{24}$$

where $D(t)$ is the active structural source defined in (18).

Compatibility between the two equations requires

$$\boxed{D(t) = -4\pi G\big(\rho_{\mathrm{str}}(t) + 3p_{\mathrm{str}}(t)\big).} \tag{25}$$

**Remark 9.1.** Equation (25) is the correct point of contact between the structural functional $D$ and the effective stress-energy sector. The source $D$ is not identified directly with $8\pi G\rho_{\mathrm{str}}$ unless an additional constitutive condition is imposed.

**Proposition 9.2** (Acceleration criterion)**.** *If*

$$D(t) > 4\pi G\big(\rho_{\mathrm{mr}}(t) + 3p_{\mathrm{mr}}(t)\big), \tag{26}$$

*then* $\ddot{a}(t) > 0$.

*Proof.* Immediate from (24). □

## 10 Constitutive closure of the structural sector

The main technical weakness of a purely formal reinterpretation is that (25) determines only the combination $\rho_{\mathrm{str}} + 3p_{\mathrm{str}}$. A constitutive closure is needed to separate $\rho_{\mathrm{str}}$ and $p_{\mathrm{str}}$.

### 10.1 General constitutive map

**Definition 10.1** (Constitutive closure map)**.** A constitutive closure map is a relation

$$\mathscr{C}\big(\rho_{\mathrm{str}}, p_{\mathrm{str}}; C, A, B, R, H, \dot{H}, \dots\big) = 0 \tag{27}$$

that, together with (25), determines the effective structural sector.

### 10.2 Provisional late-time constitutive ansatz

A disciplined late-time ansatz can be built directly from the same structural ingredients that generate $D$.

Define the burden–reserve ratio

$$\Pi(t) := \frac{B(t)}{R(t)+\varepsilon}, \tag{28}$$

and the normalised thinning fraction

$$\Upsilon(t) := \frac{\Theta[\Xi](t)}{\Theta[\Xi](t)+H(t)+\delta}, \qquad \delta > 0. \tag{29}$$

Thus $0 \le \Upsilon < 1$ whenever $H > 0$.

Now define the effective structural equation-of-state parameter by

$$\boxed{w_{\mathrm{str}}(t) = -1 + \eta_1 \frac{\dot{R}(t)}{H(t)\big(R(t)+\varepsilon\big)} - \eta_2 \frac{\dot{B}(t)}{H(t)\big(B(t)+\varepsilon\big)} - \eta_3\, \Upsilon(t),} \tag{30}$$

with constants $\eta_1, \eta_2, \eta_3 \ge 0$.

This form is not arbitrary. Each term has a clear sign interpretation:

(i) declining reserve, $\dot{R} < 0$, pushes $w_{\mathrm{str}}$ downward;

(ii) increasing burden, $\dot{B} > 0$, pushes $w_{\mathrm{str}}$ downward;

(iii) stronger coherence thinning, encoded by $\Upsilon$, pushes $w_{\mathrm{str}}$ downward.

In a late-time closure-dominant regime one therefore expects $w_{\mathrm{str}} \lesssim -1$ or $w_{\mathrm{str}} \approx -1$, with the latter recovered when the time derivatives become small and the sector approaches a quasi-static asymptote.

**Remark 10.2.** The constitutive branch of interest in this paper is the accelerating late-time branch

$$w_{\mathrm{str}}(t) < -\frac{1}{3},$$

and, in the asymptotic regime, typically $w_{\mathrm{str}}(t) \approx -1$. The line $w_{\mathrm{str}} = -1/3$ is therefore excluded as a branch boundary rather than treated as a target regime.

Given (30) and (25), one obtains

$$\boxed{\rho_{\mathrm{str}}(t) = -\frac{D(t)}{4\pi G\big(1+3w_{\mathrm{str}}(t)\big)}, \qquad p_{\mathrm{str}}(t) = w_{\mathrm{str}}(t)\rho_{\mathrm{str}}(t),} \tag{31}$$

provided $1 + 3w_{\mathrm{str}}(t) \neq 0$.

**Remark 10.3.** This is the required hardening step. The paper now contains an explicit constitutive closure, labelled provisional but mathematically usable, rather than leaving the structural sector under-determined.

### 10.3 Quasi-de Sitter limit

If

$$\dot{R} \to 0, \qquad \dot{B} \to 0, \qquad \Upsilon \to 0, \tag{32}$$

then (30) yields

$$w_{\mathrm{str}} \to -1. \tag{33}$$

In that limit (31) becomes

$$\rho_{\rm str} \to \frac{D}{8\pi G}, \qquad p_{\rm str} \to -\rho_{\rm str}. \tag{34}$$

Thus a cosmological-constant-like regime appears as a late-time structural asymptote rather than a primitive ontology.

## 11 Continuity structure and exchange bookkeeping

The homogeneous projection must satisfy the effective Bianchi identity. Define

$$\rho_{\rm eff} := \rho_{\rm mr} + \rho_{\rm str}, \qquad p_{\rm eff} := p_{\rm mr} + p_{\rm str}. \tag{35}$$

Then

$$\dot\rho_{\rm eff} + 3H(\rho_{\rm eff} + p_{\rm eff}) = 0. \tag{36}$$

If ordinary matter-radiation is separately conserved,

$$\dot\rho_{\rm mr} + 3H(\rho_{\rm mr} + p_{\rm mr}) = 0, \tag{37}$$

then the structural sector obeys

$$\dot\rho_{\rm str} + 3H(\rho_{\rm str} + p_{\rm str}) = Q(t), \tag{38}$$

where $Q$ is an effective exchange term representing unresolved transfer between the explicit homogeneous structural sector and degrees of freedom retained only implicitly by the coarse-grained projection.

This gives

$$\boxed{w_{\rm str}(t) = -1 - \frac{1}{3H}\frac{\dot\rho_{\rm str}(t) - Q(t)}{\rho_{\rm str}(t)}} \tag{39}$$

whenever $\rho_{\rm str} > 0$.

**Remark 11.1.** Equation (39) is not the primary definition of $w_{\rm str}$ in this paper. The constitutive ansatz (30) supplies the closure; (39) is then a consistency condition constraining the effective exchange term $Q$.

Combining (30), (31), and (39) yields

$$\boxed{Q(t) = \dot\rho_{\rm str}(t) + 3H\big(1 + w_{\rm str}(t)\big)\rho_{\rm str}(t).} \tag{40}$$

Thus the coarse-grained exchange sector is explicitly recoverable once the constitutive closure is chosen.

## 12 A minimal late-time structural evolution system

The paper requires a phenomenological late-time evolution system for $C, A, B, R$. This is presented as a minimal first closure, not as the final microscopic dynamics of the theory.

**Assumption 12.1** (Minimal late-time structural closure)**.** The large-scale variables satisfy

$$\dot{C} = \alpha_C HC\left(1 - \frac{C}{C_*}\right) - \sigma_C CA, \tag{41}$$

$$\dot{A} = -\alpha_A HA - \beta_A A\frac{B}{R+\varepsilon} + \gamma_A \mathcal{G}(t), \tag{42}$$

$$\dot{B} = \alpha_B HC + \beta_B A - \mu_B B, \tag{43}$$

$$\dot{R} = -\alpha_R HR - \beta_R CR - \gamma_R B + \mathcal{U}(t), \tag{44}$$

where all coefficients are nonnegative and where $\mathcal{G}(t), \mathcal{U}(t)$ are reopening or replenishment terms.

### 12.1 Status of the system

These equations are chosen for sign discipline and structural interpretation rather than for microscopic uniqueness. They encode the following minimal requirements:

(i) closure can grow under cosmic maturation but saturates;

(ii) accessible coherence is reduced by expansion and burden pressure;

(iii) burden accumulates with persistent organisation;

(iv) future reserve is compressed by expansion, closure dominance, and burden, but may be replenished by genuine reopening.

**Remark 12.2.** The role of Assumption 12.1 is limited and explicit. It provides a disciplined late-time phenomenological closure sufficient for the present paper. It is not presented as the unique structural dynamics of the full theory.

### 12.2 Pressure ratio growth

Define

$$\mathcal{P}(t) := \frac{B(t)}{R(t)+\varepsilon}. \tag{45}$$

Then

$$\dot{\mathcal{P}} = \frac{\dot{B}(R+\varepsilon) - B\dot{R}}{(R+\varepsilon)^2}. \tag{46}$$

Under the late-time sign tendencies $\dot{B} \geq 0$ and $\dot{R} \leq 0$, one has $\dot{\mathcal{P}} \geq 0$. Thus the amplification factor entering $D$ naturally grows unless reopening terms dominate.

## 13 Structural acceleration theorem

**Theorem 13.1** (Structural source criterion for late-time acceleration)**.** *Assume Assumptions 5.1 and 8.1. Let $D(t)$ be defined by (18), and let the effective homogeneous acceleration equation be (24). Then any epoch satisfying*

$$\lambda\,\Theta[\Xi](t)\,\frac{B(t)}{R(t)+\varepsilon} > 4\pi G\big(\rho_{\mathrm{mr}}(t) + 3p_{\mathrm{mr}}(t)\big) \tag{47}$$

*undergoes accelerated expansion.*

*Proof.* Substitute (18) into (24) and apply the inequality. □

**Corollary 13.2** (Primitive dark-energy ontology is not forced)**.** *The existence of an effective late-time acceleration history does not, by itself, force the ontology of a primitive dark-energy substance if a structurally coherent source satisfying the theorem exists.*

## 14 Underdetermination of ontological lift

**Definition 14.1** (Ontological lift)**.** An ontological lift of a late-time expansion history is a specification of the underlying sectors and laws taken to generate the effective homogeneous evolution.

**Proposition 14.2** (Non-uniqueness of lift)**.** *A sufficiently regular late-time expansion history does not in general determine a unique ontological lift.*

*Proof.* The homogeneous equations constrain only certain combinations of effective density and pressure. Different underlying sectors may generate the same effective combinations. In particular, a primitive-fluid lift and a structural lift may produce the same $a(t)$, $H(t)$, and effective equation-of-state history. Therefore the lift is not unique. □

**Remark 14.3.** This proposition is strictly weaker than a claim that the structural lift is correct. It states only that the standard primitive-fluid lift is not forced by expansion history alone.

## 15 Terminal closure interpretation

The reinterpretation becomes stronger when placed inside the terminal-closure architecture of Darke Theory.

The end-state picture can be summarised in the following sign chain:

(1) closure accumulates,

(2) accessible coherence thins,

(3) burden rises or remains elevated,

(4) future admissibility reserve compresses,

(5) lawful future reorganisation tends toward exhaustion.

The Darke Energy functional reads exactly this combination into the homogeneous metric sector.

**Proposition 15.1** (Late-time sign proposition)**.** *If, in a late-time regime,*

$$\Theta[\Xi](t) > 0, \qquad B(t) \text{ is large or increasing}, \qquad R(t) \text{ is small or decreasing}, \tag{48}$$

*then $D(t)$ is positive and is amplified by further reserve compression.*

*Proof.* This follows directly from (18). □

**Proposition 15.2** (Terminal-closure interpretation)**.** *Suppose the universe enters a regime in which closure accumulation suppresses the creation of new admissible future structure faster than that future structure can be replenished, and suppose the homogeneous projection remains valid. Then sustained positive $D(t)$ is the expected metric signature of progression toward terminal closure.*

*Proof.* Under the stated conditions, closure-accessible coherence thins, burden grows or remains high, and future reserve compresses. By construction, $D$ reads that combination as a positive structural source. The effective homogeneous acceleration therefore acquires a sustained positive term. The statement is interpretive but follows directly from the sign architecture of the model. □

**Remark 15.3.** The present paper intentionally states this as a proposition rather than a theorem. The claim is structurally strong, but its final empirical force depends on the success of the observational proxy programme.

## 16 Cosmological-constant-like asymptote

**Proposition 16.1** (Quasi-$\Lambda$ limit)**.** *If, on a late-time interval,*

$$\Theta[\Xi](t) \to q_\infty > 0, \qquad \frac{B(t)}{R(t)+\varepsilon} \to \Pi_\infty > 0, \qquad w_{\mathrm{str}}(t) \to -1, \tag{49}$$

*then*

$$D(t) \to D_\infty := \lambda q_\infty \Pi_\infty, \tag{50}$$

*and the structural sector approaches a cosmological-constant-like regime with*

$$\rho_{\mathrm{str}} \to \frac{D_\infty}{8\pi G}, \qquad p_{\mathrm{str}} \to -\rho_{\mathrm{str}}. \tag{51}$$

*Proof.* Immediate from (18) and (34). □

Thus a constant-like late-time sector emerges as an asymptotic structural regime rather than as a primitive vacuum ontology.

## 17 Toy late-time observational form

The present paper is a framework paper with an explicit constitutive closure, not yet a full survey-calibrated fit. It is nevertheless useful to display a simple observational form.

Assume on a late-time interval that

$$C(t) \approx C_0 e^{\alpha(t-t_0)}, \qquad \alpha \geq 0, \tag{52}$$

$$A(t) \approx A_0 e^{-\beta(t-t_0)}, \qquad \beta > 0, \tag{53}$$

$$\frac{B(t)}{R(t)+\varepsilon} \approx \Pi_0 e^{\nu(t-t_0)}, \qquad \nu \geq 0. \tag{54}$$

Then

$$\Xi(t) \approx \Xi_0 e^{-(\beta-\alpha)(t-t_0)}, \tag{55}$$

so that

$$\Theta[\Xi](t) \approx [\beta - \alpha]_+. \tag{56}$$

Hence

$$D(t) \approx \lambda[\beta - \alpha]_+ \Pi_0 e^{\nu(t-t_0)}. \tag{57}$$

In a spatially flat regime with negligible exchange and near-$-1$ late-time equation of state,

$$H^2(z) \approx H_0^2 \left[\Omega_{m0}(1+z)^3 + \Omega_{r0}(1+z)^4 + \Omega_{\rm str}(z)\right], \tag{58}$$

with

$$\Omega_{\rm str}(z) := \frac{\rho_{\rm str}(z)}{\rho_{\rm crit,0}}. \tag{59}$$

If $\rho_{\rm str}(z)$ varies slowly, the sector appears $\Lambda$-like. If it drifts, the effective dark sector acquires a mild redshift dependence.

# 18 Initial observational proxy programme

A legitimate criticism of any structural cosmology is that the variables may remain too abstract unless proxy pathways are identified. The present paper therefore records a first observational programme. These are not final estimators, and they are not claimed to be unique observables. They are classes of coarse-grained estimators intended to make the framework empirically tractable.

## 18.1 Proxy classes

**Closure density $C(z)$.** A first proxy class for closure density is the large-scale degree of persistent structural integration in the cosmic web. Candidate coarse-grained estimator classes include:

(a) mature clustering statistics at fixed scale,

(b) persistent topological connectivity measures,

(c) saturation of large-scale filamentary integration,

(d) long-lived bound structure fraction.

**Accessible coherence density $A(z)$.** A first proxy class for accessible coherence is the amount of structure still available for further lawful organisation rather than already locked into persistent mature states. Candidate coarse-grained estimator classes include:

(a) residual large-scale growth opportunity inferred from structure-growth diagnostics,

(b) variance in not-yet-saturated environments,

(c) coherence-sensitive growth residuals relative to mature regions,

(d) redshift evolution of structurally young versus mature populations.

**Persistence burden $B(z)$.** A first proxy class for burden is the effective carrying cost of historically accumulated structure. Candidate coarse-grained estimator classes include:

(a) integrated mature-structure fraction,

(b) persistence-weighted clustering measures,

(c) large-scale bound complexity load,

(d) accumulated structural lock-in indicators.

**Future admissibility reserve $R(z)$.** A first proxy class for reserve is the remaining capacity for nontrivial lawful structural novelty. Candidate coarse-grained estimator classes include:

(a) unsaturated growth reservoirs,

(b) remaining structurally underconstrained volume fraction,

(c) decline of admissible novelty indicators across redshift,

(d) coarse-grained reorganisation capacity inferred from environment distributions.

### 18.2 What the proxy programme would test

The proxy programme is not required to produce unique exact estimators at the outset. It needs only to test the core sign architecture:

$$\text{effective acceleration} \quad \longleftrightarrow \quad \text{coherence thinning} + \text{burden pressure} + \text{reserve compression}. \tag{60}$$

A successful first-stage empirical programme would therefore test whether the reconstructed late-time effective structural sector correlates with:

(i) declining accessibility of coherent future structure;

(ii) increasing persistence burden;

(iii) compression of future admissibility reserve.

## 19 Failure conditions and discriminators

The model should state plainly what would count against it.

### 19.1 Failure conditions

**F1: No workable proxies.** If no robust proxy pathway can be developed even at the coarse-grained level for $C, A, B, R$, the framework remains conceptually coherent but empirically weak.

**F2: Wrong sign structure.** If late-time effective acceleration correlates with increasing closure-accessible coherence rather than thinning, the core sign architecture is incorrect.

**F3: Burden–reserve irrelevance.** If the effective late-time sector shows no relation to increasing persistence burden or declining future reserve, the terminal-closure interpretation is weakened.

**F4: No viable constitutive calibration.** If the provisional constitutive closure cannot be improved, fitted, or replaced by a more disciplined relation, the present model remains incomplete.

**F5: Required effective history incompatible with structural signs.** If observations require a late-time sector whose sign evolution fundamentally contradicts closure-accessible coherence thinning under burden and reserve compression, the model fails in its current form.

### 19.2 Discriminators

**D1:** Reconstruction of a redshift-dependent $\rho_{\mathrm{str}}(z)$ or $w_{\mathrm{str}}(z)$ that tracks proxy estimates for thinning, burden, and reserve compression.

**D2:** Evidence that mature structural integration and decline of lawful novelty correlate with the effective acceleration sector better than a primitive constant fluid does.

**D3:** Joint modelling in which entropy, end-state, and late-time acceleration are all described within one structural lifecycle architecture rather than by disconnected auxiliary sectors.

## 20 Discussion

The paper now occupies a precise status.

It is not yet a full survey-calibrated cosmological fit. It is a hardened framework paper with an explicit constitutive closure, a disciplined effective homogeneous sector, a defined sign architecture, and a concrete proxy pathway.

Its main gain is architectural unification. Darke Theory already reinterprets geometry as derivative, gravity as a structural readability rather than a primitive force, entropy as loss of accessible coherence, and cosmic end-state as exhaustion of lawful futurehood under terminal closure. It would be structurally inconsistent to treat late-time acceleration as the sole major exception requiring an otherwise disconnected primitive dark sector.

The present paper removes that exception. It does not deny the observations. It changes the ontological reading of them.

The central distinction maintained throughout is essential:

(i) $D(t)$ is the active structural source for acceleration;

(ii) $\rho_{\mathrm{str}}(t)$ and $p_{\mathrm{str}}(t)$ are the effective homogeneous stress-energy variables;

(iii) a constitutive closure is needed to connect them.

Once that distinction is respected, the previous continuity defect disappears and the structural model becomes technically coherent.

The remaining future work is clear: derive stronger proxy estimators, fit the constitutive parameters, and test the sign architecture against late-time survey products.

## 21 Conclusion

This paper has developed a structural reinterpretation of dark energy within Darke Theory.

The central claim is that late-time accelerated expansion does not, by itself, compel the ontology of a primitive dark-energy substance, vacuum-energy density, or hidden expansive fluid. Once geometry is treated as derivative rather than primitive, an effective acceleration term in the homogeneous cosmological equations may instead be the metric appearance of deeper structural evolution.

The paper has established this in a disciplined way.

First, it placed late-time homogeneous cosmology under the relativistic closure–curvature bridge, so that FLRW-type dynamics are used only as effective post-threshold projections.

Second, it defined the relevant state variables as inherited structural quantities: closure density $C(t)$, accessible coherence density $A(t)$, persistence burden $B(t)$, and future admissibility reserve $R(t)$.

Third, it introduced the one-sided thinning operator

$$\Theta[\Xi](t) = \left[-\frac{\mathrm{d}}{\mathrm{d}t}\log \Xi(t)\right]_+, \qquad \Xi(t) = C(t)A(t),$$

thereby ensuring that positive sourcing occurs only under genuine thinning of closure-accessible coherence.

Fourth, it defined the active structural source

$$D(t) = \lambda\,\Theta[\Xi](t)\,\frac{B(t)}{R(t)+\varepsilon}$$

and embedded it consistently into the effective homogeneous equations.

Fifth, it provided a provisional constitutive closure for the effective structural sector, allowing $\rho_{\mathrm{str}}$ and $p_{\mathrm{str}}$ to be recovered from the structural variables rather than leaving the sector underdetermined.

Sixth, it linked the resulting late-time source directly to the terminal-closure architecture of Darke Theory: accumulated closure, thinning accessible coherence, rising burden, and declining future reserve.

The observational problem remains real. The acceleration data remain real. What is reinterpreted is the ontological lift from those data.

**Core structural identity.** The paper reduces to the following compact pair:

$$\boxed{D(t) = \lambda\left[-\frac{\mathrm{d}}{\mathrm{d}t}\log\left(C(t)A(t)\right)\right]_+ \frac{B(t)}{R(t)+\varepsilon}} \tag{61}$$

and

$$\boxed{\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho_{\mathrm{mr}} + 3p_{\mathrm{mr}}\right) + \frac{1}{3}D(t).} \tag{62}$$

In this form, late-time acceleration is interpreted not as evidence for a primitive hidden fluid, but as the derivative metric readability of closure-accessible coherence thinning under accumulated persistence burden and narrowing futurehood.